

# Validity Is Not Difficulty: Supplementary Material

Anonymous CVPR submission

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## A. SceneIR Schema and Coordinate Conventions

SceneIR records explicit  $T_{\text{city} \leftarrow \text{ego}}$ ,  $T_{\text{ego} \leftarrow \text{sensor}}$ , and  $T_{\text{city} \leftarrow \text{actor}}$  transforms. A return  $x_{s,t}$  is fused in Actor coordinates as

$$x_i = T_{\text{city} \leftarrow \text{actor}, t}^{-1} T_{\text{city} \leftarrow \text{ego}, t} T_{\text{ego} \leftarrow \text{sensor}, t} x_{s,t}. \quad (1)$$

The physical layer stores Actor-local surfels, known/unknown masks, normals, temporal and view support, matched rays, and source timestamps. Appearance primitives are referenced only through attachment provenance; Gaussian opacity is never read as collision occupancy. Actor identity, category, extent, trajectory, lifecycle, and analytic hazard fields are immutable across compiler actions.

**Four actions.** KEEP preserves supported query evidence. PROJECT moves an artifact only to its matched observed termination. COMPLETE adds a canonical surfel only when temporal and view support satisfy the frozen contract. UNKNOWN is the fallback when evidence is insufficient; it never becomes free.

## B. Matched-Ray Physical Certificate

Automotive LiDAR is 2.5D ray evidence, not merely an unordered 3D point set [2]. For origin  $o$ , direction  $v$ , and observed termination  $d$ , the measured free interval is

$$\mathcal{F}(o, v, d) = \{o + sv \mid 0 \leq s < d - \epsilon\}. \quad (2)$$

A provenance-bearing PROJECT candidate is placed at the same ray's termination  $o + dv$ . It therefore cannot lie in  $\mathcal{F}$ . A candidate without an exact ray is marked UNKNOWN; nearest-surface projection is not substituted. This is the source of the paired zero-free-space result. It is a deterministic contract for matched rays, not a completeness proof for the whole Actor surface or a probabilistic sensor-noise guarantee.

The held-out targets are later frames excluded from both canonical fusion and query compilation. Depth, ray-termination, Chamfer, precision, and recall are computed

against these target-only returns. This separation explains why zero paired free-space violation can coexist with the observed 17.19% per-Actor Chamfer tail.

**Bidirectional visibility partition.** For each held-out target ray, the nearest output within the frozen 0.20 m beam tube is classified as early, hit, or late; otherwise it is unmatched. In the reverse direction, an output point is supported at the first return, contradicted if it lies in the observed free interval, occluded if it is behind the return, and unknown when off ray. Occluded and unknown points are not penalized as free-space violations, preventing unseen Actor backs from being conflated with measurable contradictions.

The pooled result does not imply Actor-wise dominance: 64.04% add no early/visible contradiction and only 1.58% have absolute zero visible contradictions. All-Actor output attribution assigns 95.12% of new early rays to COMPLETE, but each costs against 14.96 new matched hits; target early returns fall by 9,439 overall. Moreover, only 13.49% of contradicted output points are completion points, so source-ray carving cannot selectively remove the observed tail. We retain completion and require the independent repair-or-abstain head for Actor-level authority.

The exact-once fresh cohort independently preserves this boundary: only 62.72% add no visible violation, 2.29% have exact zero visible contradiction, and 19.50% worsen in Chamfer. Among the latter, just 16.67% retain a non-inferior visibility F-score. Fresh confirmation therefore strengthens pooled three-dimensional evidence while continuing to reject a universal Actor certificate.

**Frozen authority–certificate alignment.** An exact identity join to the frozen P4 decisions shows why abstention must name its target. At 77.25% coverage, selected Chamfer-worsening falls from 19.50% to 14.60%, yet selected visible-failure risk is 36.39% versus 37.28% for always-repair. Its one-sided 95% Wilson upper bound is 40.40%, which remains above the baseline point risk; safe-visible AUROC is 0.533, and abstention captures 24.62% of visible failures. P4 authorizes Chamfer-nonworsening re-

Table 1. Target-nearest proximity evidence on consumed and independently frozen fresh AV2 cohorts. Higher hit/precision/F-score is better; lower early termination is better. Each audit was frozen before its metrics were read.

| Surface                     | Target-hit recall (%) | Early termination (%) | Visible precision (%) | Visibility F-score |
|-----------------------------|-----------------------|-----------------------|-----------------------|--------------------|
| Consumed clean query        | 49.69                 | 3.395                 | 99.61                 | 0.663              |
| Consumed corrupted baseline | 55.61                 | 3.589                 | 98.38                 | 0.711              |
| Consumed HARP-3D            | 69.83                 | 2.868                 | 99.77                 | 0.822              |
| Fresh clean query           | 49.72                 | 3.448                 | 99.51                 | 0.663              |
| Fresh HARP-3D               | 68.99                 | 2.931                 | 99.68                 | 0.815              |

076 pair but provides no finite-cohort confidence separation for  
077 bidirectional visibility.

078 **Why a stricter provenance rule fails.** If authority ad-  
079 ditionally requires zero COMPLETE outputs, every re-  
080 tained point has a KEEP or matched-hit PROJECT wit-  
081 ness. Provenance-only coverage is 36.90%; after conjunc-  
082 tion with P4 it is 23.71%. Dual visible failure falls to  
083 20.16% (one-sided 95% upper 26.70%), but Chamfer wors-  
084 ening rises to 43.55% and hazardous-Actor coverage is  
085 only 3.52%. Completion-count unsafe-visible AUROC is  
086 0.589. Thus observed provenance improves one physical  
087 axis while failing joint geometry and hazard authority; it  
088 cannot certify unseen views.

089 **A targeted visibility head still fails joint authority.** A  
090 frozen nuScenes-only head reaches safe-visible AUROC  
091 0.753 on source test and 0.625 on AV2. Visibility-only  
092 AV2 selection has 11.63% risk at 8.22% coverage. Its P4  
093 conjunction has 10.26% risk (95% upper 20.98%) but only  
094 7.46% coverage, 35.90% Chamfer worsening, and 3.52%  
095 hazard coverage. The ranking transfers; the authority does  
096 not.

097 **Defer-to-query changes the unit of risk.** Every ab-  
098 stained Actor retains its query surface, so Actor  
099 and hazard semantics remain 100.0%. P4's popula-  
100 tion introduced-visible failure/composite Chamfer gain is  
101 28.11%/ 0.083 m; P6-C is 30.78%/0.089 m; and P4-and-  
102 visibility is 0.76%/0.0012 m. The provenance policy is  
103 4.78%/-0.0009 m and is strictly dominated by query-only.  
104 Thus low conditional risk can coexist with a worse fallback  
105 world.

106 **Hazard-stratified accounting.** For each frozen stratum,  
107 composite gain equals coverage times selected gain, while  
108 introduced-visible mass equals coverage times selected risk  
109 (maximum numerical residual  $5.55 \times 10^{-17}$ ). Hazardous  
110 Actors are 27.15% of the cohort but account for 42.86% of  
111 P4 introduced failures and 56.42% of its Chamfer gain. P4  
112 hazard risk is 47.37% versus 46.48% for always-repair; 45  
113 of its 48 removed failures are instead clear. P6-C repairs

99.30% of hazardous Actors at 46.81% risk. Conversely,  
P4-and-visibility's zero observed hazardous failure covers  
only 3.52% of hazardous Actors. Thus hazard-state reten-  
tion is not a hazard-stratum visibility certificate.

**Fresh target-nearest action provenance separates two failure channels.** Across 1,435,391 target rays, the com-  
piled surface adds 1.348% proxy new-early returns and  
yields 14.51 proxy new hits per proxy new-early return.  
COMPLETE accounts for 94.70% of proxy new early re-  
turns and 99.94% of proxy new hits, whereas KEEP ac-  
counts for 95.68% of emitted-surface contradictions. Haz-  
ardous/clear proxy new-early rates are 1.912%/0.980%;  
their COMPLETE hit-to-early ratios are 13.67/17.62. P4  
and P6-C hazardous rates remain  $1.006 \times / 1.001 \times$  always-  
repair. Thus the selectors do not suppress the completion-  
driven hazardous-ray mechanism. PROJECT has zero emit-  
ted provenance because observed-hit projections collapse  
under KEEP-first voxel deduplication; this is not a causal  
zero-effect result.

### C. Literal First-Return Correction and Mono- tone Boundary

The frozen AV2 attribution above selects the Euclidean-  
nearest output to each target point and then checks its pro-  
jected depth. It is a target-aligned proximity diagnostic, not  
literal ray termination. Following raw-LiDAR occupancy  
evaluation [3], P20/P22/P23 instead take the minimum po-  
sitive projected depth among every surface point inside the  
same frozen beam tube. Query and compiled surfaces use  
the identical operator.

$$d_S(r) = \min\{s > 0 : o + sv \in S_e\}, \quad \min \emptyset = \infty. \quad (3)$$

**Proposition 1 (one-sided deletion certificate).** Fix ray  $r$ , tube radius  $\epsilon$ , target depth  $d^*$ , tolerance  $\tau$ , and query surface  $Q$ . If  $S' \subseteq S$ , then  $d_{S'}(r) \geq d_S(r)$ . Consequently, both the early indicator and new-early relative to  $Q$  are non-increasing:

$$e_Q(S; r) = \mathbf{1}[d_S(r) < d^* - \tau] \mathbf{1}[d_Q(r) \geq d^* - \tau], \quad (4)$$

$$e_Q(S'; r) \leq e_Q(S; r).$$

*Proof.* The feasible positive depths in  $S'$  are a subset of those in  $S$ ; the minimum therefore cannot decrease, including the empty-set case  $\min \emptyset = \infty$ . Thresholding by the fixed  $d^* - \tau$  preserves the order, and the query factor is constant.  $\square$

The premise covers deletion only. Projection, completion, point motion, ray/tolerance changes, and unobserved collision queries lie outside it. Even under deletion, the sole point supporting a target-depth hit may be removed, while the two directed terms of symmetric Chamfer need not share a direction. Table 2 shows this exact boundary on the consumed nuScenes source-development cohort: all three frozen deletion policies reduce hazardous early returns, yet all pay surface utility. P19 is most efficient, removing 131 hazardous events for 71 hits and 0.0919 mm mean Chamfer (1.845 events/hit; 1426.1 events/mm). These ratios are descriptive, not collision or road-safety guarantees.

P22 applies the identical frozen operator to the already consumed AV2 cohort rather than relabeling the proxy. Across 523 Actors and 1,435,391 target rays, all/hazard/clear new-early rates change from 1.348/1.912/0.980% to 9.894/13.926/7.256%, or  $7.34/7.29/7.41\times$ . Every one of the 142,022 literal new-early returns has COMPLETE provenance; completion also supplies 97.843% of literal new hits, yielding only 1.194 hits per early return. This corrects, rather than confirms, the larger target-nearest efficiency ratio reported above.

P23 prospectively freezes the same comparison on a disjoint 10-log AV2 cohort before any quality read. Across 233 Actors and 650,145 rays, all/hazard/clear rates change from 1.370/1.788/0.948% to 10.801/14.817/6.750%, or  $7.89/8.29/7.12\times$ . All 70,220 literal new-early returns are COMPLETE, and the hit/early ratio remains only 1.237. This is fresh confirmation of metric undercounting and provenance, not of a model or policy.

## D. Frozen Cohorts and Exact-Once Rules

After a quality read, scenes/logs cannot be replaced, failed cases remain in the denominator, and no threshold, radius, feature group, or checkpoint may change. Download/resource failures are reported separately from scientific failures. The fresh AV2 protocol is bound to the already frozen P6-C/P4 artifacts and is not changed by the rejected fresh nuScenes result.

The fresh AV2 read contains 523 Actors. P6-C/P4 repair AUROC is 67.62/65.48, coverage is 83.94/77.25%, and selective Chamfer is 0.178/0.184 m. This external reversal does not dominate P4: false repair is 12.43% versus 11.28%, and P6-C independently loses fresh-nuScenes AUROC and AURC. The complete evidence therefore retains P4 and rejects a domain-invariant sparsity-consistency claim.

## E. Actor-Level Decision Intervals

For the frozen ReLU validity MLP, an input box  $[\ell, u]$  is propagated through each affine layer using

$$\ell' = W^+ \ell + W^- u + b, \quad u' = W^+ u + W^- \ell + b, \quad (5)$$

followed by endpoint-wise ReLU. Here  $W^+ = \max(W, 0)$  and  $W^- = \min(W, 0)$ . The resulting logit interval is mapped through the sigmoid. An Actor is *robust-select* if its lower probability is above the frozen P4 threshold, *robust-abstain* if its upper probability is below threshold, and unresolved otherwise. This is deterministic interval-bound propagation over a specified box, related to robust-OOD certification [1]; it does not assert that the box describes an unseen sensor.

We fix radii 0.05/0.10/0.20 in nuScenes-training standard deviations. The sensor-opportunity group contains point/surfel counts, temporal/view support, range, and frame count; the physical-surface group contains surface residuals, keep/unknown fractions, completion fraction, and compiled/query ratio. Their union is exactly the 13D validity input.

At radius 0.10, 33.33% of nominal nuScenes-test selections and 91.23% of consumed-AV2 selections are robust-select. Nevertheless, 10.76% of the latter are retained target false repairs. Query-point count is the largest one-feature interval source for 83.33%/95.11% of nuScenes/AV2 Actors. Thus a shifted model can be confidently and locally stable while wrong.

## F. Intended Use and Safety Boundary

HARP-3D is intended for offline reconstruction audits: expose ray contradictions, compare Actor-preserving repairs, and report separate empirical task reliability. Consumers must retain unknown/provenance state and defer; repair never authorizes control. It is not a production collision map, closed-loop certificate or controller, or safety-case evidence. False repair can change collision support, sensor shift can cause confident errors, and selection can hide hazardous-Actor coverage. Deployment requires measured sensor errors, independent map/dynamics checks, closed-loop evaluation, and a domain-specific safety process. The interfaces expose, but do not remove, residual risk.

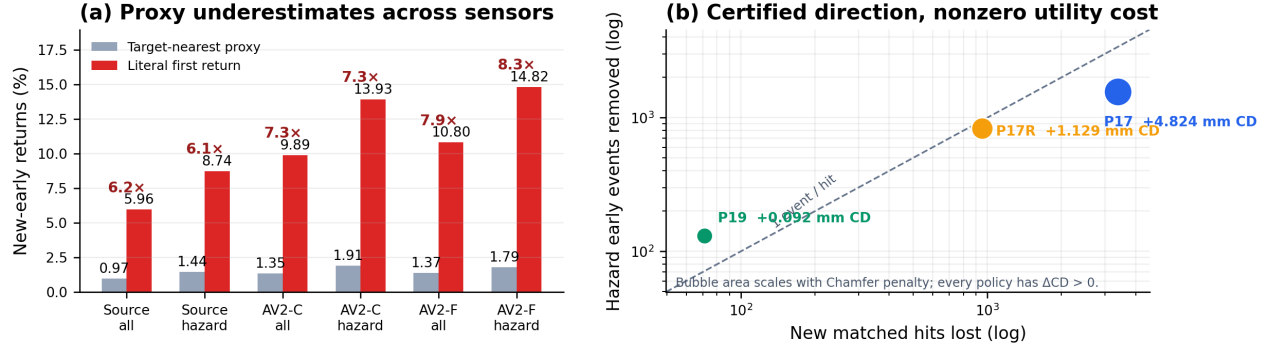


Figure 1. Frozen literal first-return boundary. (a) The target-nearest proxy understates total and hazardous-Actor new-early exposure on source, consumed AV2 (AV2-C), and exact-once fresh AV2 (AV2-F). (b) Deleting points is directionally safe for early returns, but every frozen source policy loses matched hits and worsens Chamfer. Bubble area scales with the Chamfer penalty. Ratios are consumed-source diagnostics; fresh AV2 transfers the metric failure only, not a model, policy, or road-safety bound.

Table 2. Literal first-return source audit. Hazard early is the fraction of held-out hazardous-Actor rays newly terminated before the target. Chamfer and new hits use the same frozen surfaces; no policy is refit.

| Surface policy    | Completion coverage (%) | Hazard new-early (%) | Mean Chamfer (m) | New hits |
|-------------------|-------------------------|----------------------|------------------|----------|
| Always-complete   | 100.00                  | 8.742                | 0.194587         | 32,575   |
| P17 ray-only      | 71.43                   | 7.777                | 0.199411         | 29,191   |
| P17R ray+Chamfer  | 88.96                   | 8.230                | 0.195716         | 31,620   |
| P19 one-slot veto | 98.95                   | 8.662                | 0.194679         | 32,504   |

Table 3. Cohort construction is metadata-only. No quality, model score, RGB appearance, or target metric is read before freezing.

| Cohort                      | Size      | Frozen rule                                                                                                                                         |
|-----------------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| AV2 consumed external       | 30 logs   | Sort all 150 Sensor-val UUIDs; take indices 0, 5, ..., 145. First 20 are quantitative and final 10 qualitative.                                     |
| AV2 fresh exact-once        | 20 logs   | Remove the consumed indices; from the 120-log complement take positions 0, 6, ..., 114.                                                             |
| AV2 fresh metric exact-once | 10 logs   | Remove both prior AV2 cohorts; from the 100-log complement take positions 0, 10, ..., 90, frozen before P22.                                        |
| nuScenes fresh exact-once   | 20 scenes | Exclude every P4 train/calibration/test scene; sort the remaining 106 available scenes by official index and take rounded equally spaced positions. |

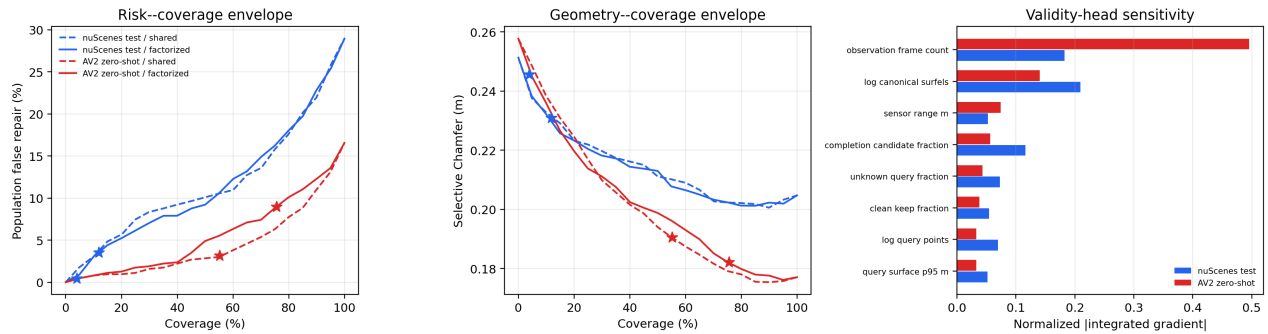


Figure 2. Complete frozen P7 explanation. Left and center: risk-coverage and geometry-coverage curves; stars mark the unchanged nuScenes-calibrated thresholds and never select an AV2 operating point. Right: Integrated Gradients relative to the nuScenes-training mean. The AV2 shift toward observation-frame count exposes sensor-opportunity dependence. The curves and attributions are descriptive, not a cross-domain certificate.

## G. Failure Ledger and Claim Corrections

Table 4. Paper-facing failures are retained as boundaries rather than hidden by replacement experiments.

| ID     | Observation                                                                                                              | Resulting boundary or correction                                                                                                             |
|--------|--------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| V7-F09 | Aggregate geometry hides 17.19% Actor-level Chamfer worsening.                                                           | P3 is aggregate hard evidence; P4 adds repair-or-abstain and preserves all Actors.                                                           |
| V7-F11 | AV2 validity attribution is dominated by sensor opportunity.                                                             | Zero hazard leakage does not imply domain-invariant validity.                                                                                |
| V7-F13 | Exact P4-train overlap is only two scenes/five Actors.                                                                   | Reject joint fitting; audit the frozen physical/reliability interface only.                                                                  |
| V7-F14 | FP32 exceeds the analytic tolerance by $3.8 \times 10^{-6}$ .                                                            | Recompute the same bound in FP64; do not loosen the tolerance or alter rows.                                                                 |
| V7-F15 | Fresh P6-C repair AUROC loses 3.50 points to P4.                                                                         | Reject recovery despite higher operating-point coverage; P4 remains primary.                                                                 |
| V7-F16 | Freeze prose mislabeled the P346 held-out horizon.                                                                       | Executable artifact is 2.5 s; correct documentation without rerunning.                                                                       |
| V7-F17 | 47 AV2 false repairs remain robust-select at radius 0.10.                                                                | Network stability is not correctness, calibration, or a safety certificate.                                                                  |
| V7-F18 | Pooled ray physics improves, but consumed/fresh nonnew-visible rates are only 64.04/62.72%.                              | Retain aggregate C1 evidence and the completion gain-tail trade-off; reject a universal Actor certificate and do not tune the ray tolerance. |
| V7-F19 | P6-C loses fresh-nuScenes ranking but improves fresh-AV2 AUROC while raising false repair.                               | Report the domain reversal; an external gate pass neither overrides the source rejection nor promotes P6-C over P4.                          |
| V7-F20 | P4 visible risk is 36.39%, but its 40.40% upper bound exceeds the 37.28% baseline.                                       | Repairability does not confidence-separate visibility risk.                                                                                  |
| V7-F21 | Provenance lowers visible risk but raises Chamfer worsening to 43.55% and hazard-repair coverage falls to 3.52%.         | Observed provenance is one-axis evidence, not joint authority.                                                                               |
| V7-F22 | The source visibility head transfers ranking, but dual coverage/gain are negligible and hazard-repair coverage is 3.52%. | Reject the head family rather than sweep target thresholds.                                                                                  |
| V7-F23 | Provenance defer has -0.0009 m composite gain with nonzero introduced failure.                                           | Query-only strictly dominates it; conditional risk is not fallback-system utility.                                                           |
| V7-F29 | Literal first-return raises baseline hazard exposure to 8.742%; P17/P17R/P19 all lower it but worsen Chamfer.            | Point deletion has a certified early-return direction, not hit, Chamfer, collision, or policy safety.                                        |

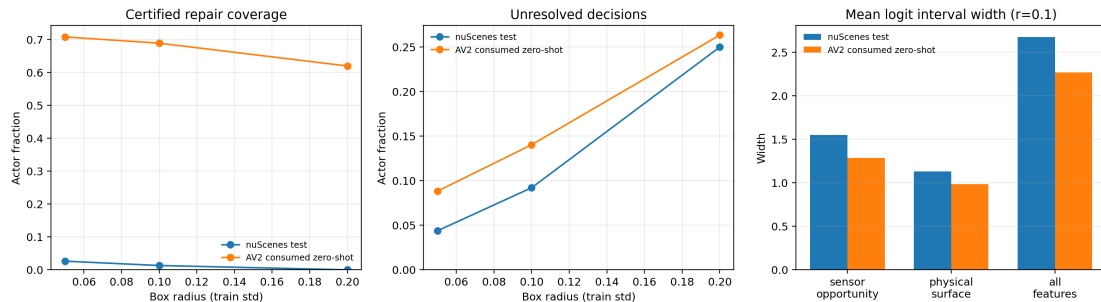


Figure 3. Frozen P7-C interval audit. Left/center: robust-select coverage and unresolved fraction across pre-registered radii. Right: sensor-opportunity boxes induce wider logit intervals than physical-surface boxes. The high AV2 certified coverage does not erase the 47 stable false repairs.



## 243 H. Frozen AV2 Camera-Evidence Gallery

244 The following ten cases are the actor-rank-0 case from each qualitative log in the pre-registered order, not a metric or  
245 appearance ranking. Every panel shows synchronized RGB, the paired synthetic artifact probe, the four-action physical  
246 output, and sparse camera depth. The full package contains 30 panels and 30 before/after videos (three lexically first eligible  
247 Actors per log). Videos diagnose temporal behavior but are not photorealistic renders.



Figure 4. Frozen case q00, hazardous Actor 1f197615, ring-front-center; 251 visible query returns and 2,287 sparse-depth points in crop.



Figure 5. Frozen case q01, non-hazardous Actor 0e0562b6, ring-rear-left; 137 visible query returns and 1,770 sparse-depth points.



Figure 6. Frozen case q02, non-hazardous Actor 08fc423e, ring-side-right; 85 visible query returns and 2,476 sparse-depth points.



Figure 7. Frozen case q03, hazardous Actor 05cccb41, ring-side-left; 2,456 visible query returns and 17,286 sparse-depth points.

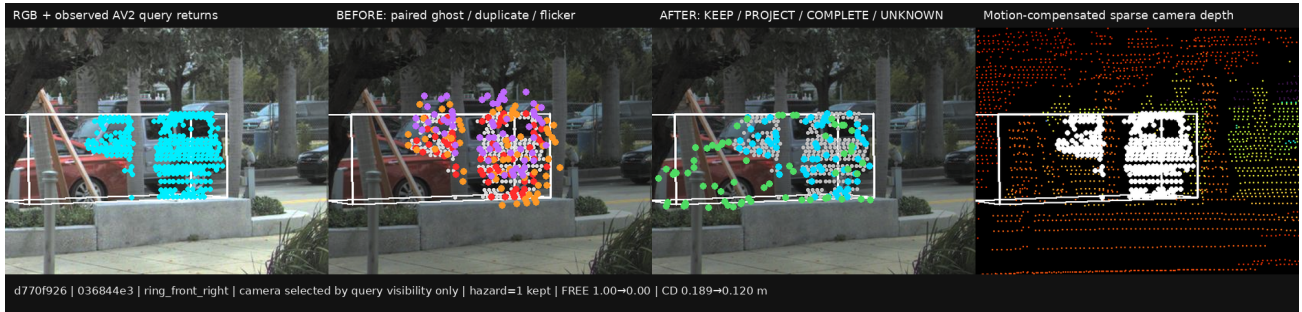


Figure 8. Frozen case q04, hazardous Actor 036844e3, ring-front-right; 292 visible query returns and 2,358 sparse-depth points.

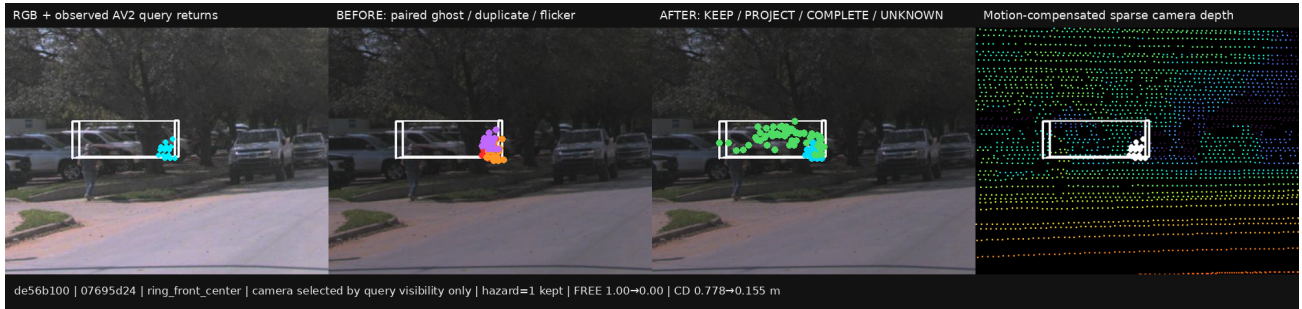


Figure 9. Frozen case q05, hazardous far/sparse Actor 07695d24, ring-front-center; all 14 query returns remain visible.



Figure 10. Frozen case q06, non-hazardous Actor 08515e0d, ring-front-right; 79 visible query returns and 2,757 sparse-depth points.





Figure 11. Frozen case q07, non-hazardous Actor 076c9103, ring-front-center. This unfiltered case exposes Chamfer worsening and is retained.

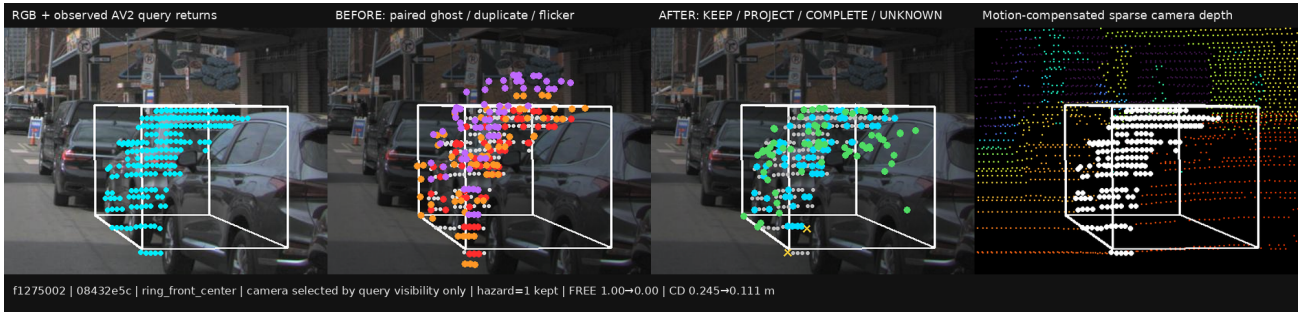


Figure 12. Frozen case q08, hazardous Actor 08432e5c, ring-front-center; 231 visible query returns and 2,101 sparse-depth points.



Figure 13. Frozen case q09, non-hazardous Actor 6d55ff95, ring-front-center; 38 visible query returns and 1,630 sparse-depth points.



## I. Resource and Reproducibility Boundary

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All V7 scientific runs use one RTX 3090. The largest paper-facing GPU allocation is below 1.5 GiB, while P7-C uses 9.2 MiB and 1.37 s. Download and preprocessing are sequential on the low-memory host. Artifacts are identified by task/run paths and immutable cohort/config files; the protocol intentionally does not add hashes, checksums, fingerprints, repeated smoke matrices, or post-hoc quality gates.

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## References

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- [1] Julian Bitterwolf, Alexander Meinke, and Matthias Hein. Certifiably adversarially robust detection of out-of-distribution data. In *Advances in Neural Information Processing Systems*, 2020. 3
- [2] Peiyun Hu, Jason Ziglar, David Held, and Deva Ramanan. What you see is what you get: Exploiting visibility for 3D object detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 11001–11009, 2020. 1
- [3] Jonas Kälble, Sascha Wirges, Maxim Tatarchenko, and Eddy Ilg. Accurate training data for occupancy map prediction in automated driving using evidence theory. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 5281–5290, 2024. 2

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